Lecture 7

Dynamics: Stone's theorem and the free wave packet

Objectives. *Stone's theorem says that the generator of every strongly continuous unitary group is self-adjoint. We prove this with vector integrals. The converse is proved for explicit diagonal and multiplication operators and sketched in general. For a free state with a rapidly decreasing transform, we compute the position variance in time.*

- Identify the generator domain from difference quotients.
- Explain why the Dirichlet momentum generates no unitary group.
- Compute free-packet spreading when $\tilde{\psi} \in \mathcal{S}(\mathbb{R})$.

Lecture 7, first part

Unitary groups and Stone's theorem

7.1 Unitary groups and their generators

In *Quantum Mechanics I*, Thm 4.4, and in Theorem 2.2, the evolution of a system is a family of unitaries U_t with $U_0 = \mathbb{1}$ and $U_{t+s} = U_t U_s$, built from an eigenbasis of the Hamiltonian. The free energy on the line, like the position and the momentum, has no eigenbasis (Theorem 4.6). This lecture starts from the evolution instead.

We assume that the evolution is reversible and linear and that it preserves inner products. Then each U_t is unitary. We also assume that the law of motion does not change in time. Then evolving for a time s and then for a time t is evolving for the time $t + s$, and evolving for no time changes nothing. This is the group law. Finally, we assume that no state jumps: for each vector ψ, the vector $U_t\psi$ changes continuously with t.

Definition 7.1 (Unitary group)**.** A *strongly continuous unitary group* on $\mathcal{H}$ is a family $(U_t)_{t\in\mathbb{R}}$ of unitary operators such that

$$U_0 = \mathbb{1} \quad \text{and} \quad U_{t+s} = U_t U_s \quad \text{for all real } t \text{ and } s,$$

and such that, for every $\psi \in \mathcal{H}$, the map $t \mapsto U_t\psi$ is continuous in norm.

Since $U_t U_{-t} = U_0 = \mathbb{1}$, the inverse of U_t is U_{-t}, and since U_t is unitary, $U_t^\dagger = U_{-t}$. By Theorem 2.2, the evolution of the box is such a group.

Continuity in the operator norm would be a stronger requirement, and the box fails it. At the times $t_n = \pi\hbar/E_n$, which tend to 0, the phase of u_n is $e^{-i\pi} = -1$, so $U_{t_n}u_n = -u_n$ and

$$\|U_{t_n} - \mathbb{1}\| \geq \|(U_{t_n} - \mathbb{1})u_n\| = 2.$$

So the definition asks for continuity on each vector separately.

Definition 7.2 (Generator)**.** Let (U_t) be a strongly continuous unitary group. Its *generator* K has the domain $D(K)$ of the vectors ψ for which $(U_h\psi - \psi)/h$ has a limit in norm as $h \to 0$, and the action

$$K\psi = \mathrm{i} \lim_{h\to 0} \frac{U_h\psi - \psi}{h}.$$

The *Hamiltonian* of the group is $H = \hbar K$.

The generator has the units of a frequency, an inverse time, and $\hbar$ turns it into an energy. For the box, Theorem 2.2 says that $(U_h\psi - \psi)/h$ converges exactly when $\psi \in D(H)$, and then to $-(\mathrm{i}/\hbar)H\psi$. So the generator of the box's group is $H/\hbar$, and its Hamiltonian is the box Hamiltonian H. The factor i and the sign follow the convention $U_t = e^{-\mathrm{i}tH/\hbar}$. Hall and Moretti write the group as $e^{\mathrm{i}tA}$, so their generator A is our $-K$.

Proposition 7.3 (What a generator does)**.** *Let (U_t) be a strongly continuous unitary group with the generator K.*

1. *The domain $D(K)$ is a subspace, and K is linear on it. Each U_t maps $D(K)$ into $D(K)$, and $KU_t\psi = U_tK\psi$ for $\psi \in D(K)$.*
2. *If $\psi \in D(K)$, then $t \mapsto U_t\psi$ is differentiable in norm at every t, and*

$$\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}t}U_t\psi = KU_t\psi. \tag{7.1}$$

   If $\psi \notin D(K)$, then $t \mapsto U_t\psi$ is differentiable at no t.
3. *For all $\varphi, \psi \in D(K)$, $\langle \varphi, K\psi\rangle = \langle K\varphi, \psi\rangle$.*

Proof.

1. A sum of two convergent quotients converges, so $D(K)$ is a subspace and K is linear. Let $\psi \in D(K)$. Since
$$\frac{U_hU_t\psi - U_t\psi}{h} = U_t\,\frac{U_h\psi - \psi}{h}$$
and U_t is bounded, the left side tends to $U_t(-\mathrm{i}K\psi)$. So $U_t\psi \in D(K)$, and $KU_t\psi = U_tK\psi$.
2. By the same identity, $(U_{t+h}\psi - U_t\psi)/h$ tends to $-\mathrm{i}U_tK\psi = -\mathrm{i}KU_t\psi$. Conversely, suppose that $(U_{t+h}\psi - U_t\psi)/h$ tends to a vector χ. Applying the bounded operator U_{-t} shows that $(U_h\psi - \psi)/h$ tends to $U_{-t}\chi$, so $\psi \in D(K)$.
3. Since $U_h^\dagger = U_{-h}$ and the inner product is antilinear in its first slot,
$$\left\langle \varphi, \mathrm{i}\,\frac{U_h\psi - \psi}{h}\right\rangle = \left\langle \mathrm{i}\,\frac{U_{-h}\varphi - \varphi}{-h}, \psi\right\rangle.$$
As $h \to 0$, the left side tends to $\langle \varphi, K\psi\rangle$, and the right side, with $-h$ in place of h, to $\langle K\varphi, \psi\rangle$. ∎

With $H = \hbar K$, the equation (7.1) is the Schrödinger equation $\mathrm{i}\hbar\,\mathrm{d}U_t\psi/\mathrm{d}t = HU_t\psi$. It holds exactly for the states in the domain of H, as in the box, and every state still evolves. The last claim says that K is symmetric once $D(K)$ is known to be dense (Definition 0.3). § 7.3 proves that it is.

Self-check 7.1

1. Let c be real, and let $U_t = e^{-\mathrm{i}ct}\mathbb{1}$. Show that (U_t) is a strongly continuous unitary group, and find its generator and the domain of the generator.

7.2 Evolution by an explicit Hamiltonian

Every operator that the course has diagonalised generates a group. As in Proposition 6.8, these are of two kinds.

Theorem 7.4 (Evolution by an explicit Hamiltonian)**.** *Let B be one of the following operators.*

1. Diagonal. *For an orthonormal basis (e_k) and real numbers λ_k, with $c_k = \langle e_k, \psi\rangle$,*

$$D(B) = \Big\{\psi : \sum_k \lambda_k^2 |c_k|^2 < \infty\Big\}, \qquad B\psi = \sum_k \lambda_k c_k e_k.$$

   Put $e^{-\mathrm{i}tB/\hbar}\psi = \sum_k e^{-\mathrm{i}t\lambda_k/\hbar} c_k e_k$.
2. Multiplication. *For a real continuous function g on $\mathbb{R}$, with M_g as in Lemma 0.7, and a unitary U, $B = U^{-1}M_gU$. Put $e^{-\mathrm{i}tB/\hbar} = U^{-1}M_{e^{-\mathrm{i}tg/\hbar}}U$.*

Then $(e^{-\mathrm{i}tB/\hbar})_{t\in\mathbb{R}}$ is a strongly continuous unitary group, and its generator is $B/\hbar$, with the domain $D(B)$.

Proof. First, the group, and its derivative on $D(B)$. For a diagonal B, the proof of Theorem 2.2 applies word for word, with e_k and λ_k in place of u_n and E_n, and with $|\lambda_k|$ in place of E_n in its bounds. It shows that the operators $e^{-\mathrm{i}tB/\hbar}$ form a strongly continuous unitary group, and that $(e^{-\mathrm{i}hB/\hbar}\psi - \psi)/h \to -(\mathrm{i}/\hbar)B\psi$ for $\psi \in D(B)$.

For $B = U^{-1}M_gU$, the unitary U^{-1} carries the group, its continuity and its derivatives over from $L^2(\mathbb{R})$, and $D(B) = U^{-1}D(M_g)$. So let $U = \mathbb{1}$, and write M_t for multiplication by $e^{-\mathrm{i}tg/\hbar}$. It is unitary, since $\left|e^{-\mathrm{i}tg/\hbar}\right| = 1$, and the group law holds at each point. Two bounds on real numbers θ do the rest:

$$\left|e^{-\mathrm{i}\theta} - 1\right| \le |\theta| \quad \text{and} \quad \left|e^{-\mathrm{i}\theta} - 1 + \mathrm{i}\theta\right| \le \frac{\theta^2}{2}, \tag{7.2}$$

since $e^{-\mathrm{i}\theta} - 1 = -\mathrm{i}\int_0^\theta e^{-\mathrm{i}u}\,\mathrm{d}u$ and $e^{-\mathrm{i}\theta} - 1 + \mathrm{i}\theta = -\int_0^\theta\int_0^u e^{-\mathrm{i}v}\,\mathrm{d}v\,\mathrm{d}u$, whose integrands have modulus 1. For continuity, let $\varphi \in L^2(\mathbb{R})$ and $\epsilon > 0$. By the continuity clause of Definition 5.1 for the spectral measure of Q (Proposition 5.3), there is R with $\left\|\varphi - 1_{[-R,R]}\varphi\right\|^2 < \epsilon$. Since g is continuous, $|g| \le G$ on $[-R,R]$ for a constant G. On $[-R,R]$ the first bound applies, and outside it $\left|e^{-\mathrm{i}tg/\hbar} - 1\right|^2 \le 4$, so

$$\|M_t\varphi - \varphi\|^2 = \int_{-\infty}^{\infty} \left|e^{-\mathrm{i}tg/\hbar} - 1\right|^2 |\varphi|^2\,\mathrm{d}x \le \left(\frac{tG}{\hbar}\right)^2 \|\varphi\|^2 + 4\epsilon,$$

which is less than 5ϵ for small $|t|$. Continuity at every t follows, since $M_{t+h}\varphi - M_t\varphi = M_t(M_h\varphi - \varphi)$ and M_t keeps norms. For the derivative, the two bounds with $\theta = hg(x)/\hbar$ give

$$\left|\frac{e^{-\mathrm{i}hg/\hbar} - 1}{h} + \frac{\mathrm{i}g}{\hbar}\right| \le \min\Big(\frac{2\,|g|}{\hbar},\ \frac{|h|\,g^2}{2\hbar^2}\Big).$$

Let $\varphi \in D(M_g)$, and choose R with $\left\|g\varphi - 1_{[-R,R]}g\varphi\right\|^2 < \epsilon$, and G as before. The second entry of the minimum applies on $[-R,R]$, and the first outside it, so

$$\|(M_h\varphi - \varphi)/h + (\mathrm{i}/\hbar)g\varphi\|^2 \le (hG^2/2\hbar^2)^2\,\|\varphi\|^2 + 4\epsilon/\hbar^2.$$

Hence $(M_h\varphi - \varphi)/h \to -(\mathrm{i}/\hbar)g\varphi$.

Next, the generator K. By the first step, K extends $B/\hbar$. The operator B is self-adjoint, by Proposition 5.5 or by Lemmas 0.7 and 0.8, so $D(B)$ is dense, and so is $D(K)$. By Proposition 7.3, K is symmetric, so $K \subseteq K^\dagger$. Since adjoints reverse inclusions (Proposition 0.5), $K^\dagger \subseteq (B/\hbar)^\dagger = B/\hbar$. Hence $B/\hbar \subseteq K \subseteq K^\dagger \subseteq B/\hbar$, and $K = B/\hbar$. ∎

This covers the diagonal operators of the course, namely the box Hamiltonian (Lecture 2), the circle's momentum and, by Theorem 3.10, each P_θ. With $U = \mathbb{1}$ or $U = F$ and $g(x) = x$, $g(p) = p$ or $g(p) = p^2/2m$, it covers the operators Q, P and H_0 of Theorem 4.6. For the free particle,

$$U_t\psi = e^{-\mathrm{i}tH_0/\hbar}\psi = F^{-1}\big(e^{-\mathrm{i}tp^2/2m\hbar}\tilde{\psi}\big). \tag{7.3}$$

The phase has modulus 1, so the momentum density $\big|(\widetilde{U_t\psi})\big|^2 = |\tilde{\psi}|^2$ never changes. The packet spreads, as § 7.4 shows, but it gains no momentum. Spreading is not a disturbance.

The general case needs the spectral theorem.

Theorem 7.5 (Every self-adjoint Hamiltonian evolves · *sketched*)**.** *Let H be self-adjoint. There is a strongly continuous unitary group $(e^{-\mathrm{i}tH/\hbar})_{t\in\mathbb{R}}$ whose generator is $H/\hbar$.*

Proof sketch. By the spectral theorem in Hall's multiplication form, Thm 10.10, cited, there are a measure space (X, μ), a real measurable function g on X and a unitary $U : \mathcal{H} \to L^2(X, \mu)$ with $H = U^{-1}M_gU$. The proof of the multiplication case of Theorem 7.4 applies on X, with the sets where $|g| \le G$ in place of $[-R, R]$. The step that makes the remainders small now needs the continuity of the measure μ, a cited tool. ∎

Hall, Prop. 10.14, Teschl, Thm 5.1, and Amrein, §5.1.1, give this proof with the spectral measure. By Hall's functional calculus, Def. 10.5, used for the group in his Prop. 10.14 with the opposite sign, the group is $\int e^{-\mathrm{i}t\lambda/\hbar}\,\mathrm{d}E_H(\lambda)$. By Proposition 7.9 below, the group does not depend on the choice of U and g, and for the operators of Theorem 7.4 it is the group built there.

Self-check 7.2

1. Let $\tilde{\psi}(p) = 1/(1+p^2)$. Is ψ in $D(P)$? In $D(H_0)$? At which times is the free evolution $t \mapsto U_t\psi$ differentiable?

7.3 Stone's theorem

By Proposition 7.3, the generator of a strongly continuous unitary group satisfies the symmetry identity. Stone's theorem says that the generator is self-adjoint. The proof builds vectors in $D(K)$ by averaging $U_t\varphi$ over t, with the weight $e^{-t/\tau}$. First, the average over the future lies in $D(K)$, and this shows that $K - \mathrm{i}/\tau$ maps $D(K)$ onto $\mathcal{H}$. Next, as $\tau \to 0$ the normalised average tends to φ, so $D(K)$ is dense. Then, the average over the past does the same for $K + \mathrm{i}/\tau$. Finally, the test of Proposition 6.7 gives that K is self-adjoint. These averages are integrals of vector functions, which we define first.

Definition 7.6 (Integral of a continuous vector function)**.** Let $f : [a, b] \to \mathcal{H}$ be continuous in norm. A partition $a = t_0 < t_1 < \cdots < t_N = b$ has the *mesh* $\max_j(t_j - t_{j-1})$ and the *Riemann sum* $\sum_j f(t_j)(t_j - t_{j-1})$. The *integral* $\int_a^b f(t)\,\mathrm{d}t$ is the limit of the Riemann sums as the mesh tends to 0, and $\int_b^a f(t)\,\mathrm{d}t = -\int_a^b f(t)\,\mathrm{d}t$.

Lemma 7.7 (Vector integrals)**.** *The limit exists. It is linear in f, it adds over adjacent intervals, and a shift of the variable does not change it: $\int_a^b f(t+c)\,\mathrm{d}t = \int_{a+c}^{b+c} f(s)\,\mathrm{d}s$. For every*

bounded operator A, $A\int_a^b f(t)\,\mathrm{d}t = \int_a^b Af(t)\,\mathrm{d}t$. Moreover $\left\|\int_a^b f(t)\,\mathrm{d}t\right\| \le \int_a^b \|f(t)\|\,\mathrm{d}t$, and

$$\frac{1}{h}\int_t^{t+h} f(s)\,\mathrm{d}s \longrightarrow f(t) \quad \text{as } h \to 0,$$

for t and $t+h$ in $[a,b]$.

Proof. As for real functions, f is uniformly continuous on $[a,b]$. Given $\epsilon > 0$, there is $\delta > 0$ with $\|f(s) - f(t)\| < \epsilon$ for $|s-t| < \delta$. Let a partition have mesh less than δ, and let a second partition contain all its points. Each interval of the first splits into intervals of the second, and the terms on them differ by values of f at points less than δ apart. So the two Riemann sums differ by at most $\epsilon(b-a)$. Two partitions of mesh less than δ have such a common refinement, so their sums differ by at most $2\epsilon(b-a)$. Hence the sums along partitions with mesh tending to 0 form a Cauchy sequence, and since $\mathcal{H}$ is complete, they converge, all to the same limit. The other claims hold for each Riemann sum, by the triangle inequality for the norm bound, and they pass to the limit since A and the norm are continuous. Finally,

$$\left\|\frac{1}{h}\int_t^{t+h} f(s)\,\mathrm{d}s - f(t)\right\| = \left\|\frac{1}{h}\int_t^{t+h}\big(f(s)-f(t)\big)\mathrm{d}s\right\| \le \max_{|s-t|\le|h|}\|f(s)-f(t)\| \longrightarrow 0. \qquad \blacksquare$$

The proof of Stone's theorem needs one integral over $[0,\infty)$. Let $f : [0,\infty) \to \mathcal{H}$ be continuous, with $\|f(t)\| \le Ce^{-t/\tau}$ for constants C and $\tau > 0$. For $T < T'$, the norm bound gives

$$\left\|\int_T^{T'} f\right\| \le \int_T^{T'} Ce^{-t/\tau}\,\mathrm{d}t \le C\tau e^{-T/\tau},$$

so the integrals $\int_0^T f$ form a Cauchy family as $T \to \infty$. Their limit is $\int_0^\infty f(t)\,\mathrm{d}t$, and it keeps the properties of Lemma 7.7.

Theorem 7.8 (Stone's theorem). *Let (U_t) be a strongly continuous unitary group with the generator K. Then $D(K)$ is dense, and K is self-adjoint. For every time $\tau > 0$ and every $\varphi \in \mathcal{H}$,*

$$(K - \mathrm{i}/\tau)^{-1}\varphi = \mathrm{i}\int_0^\infty e^{-t/\tau}\,U_t\varphi\,\mathrm{d}t \quad \text{and} \quad (K+\mathrm{i}/\tau)^{-1}\varphi = -\mathrm{i}\int_0^\infty e^{-t/\tau}\,U_{-t}\varphi\,\mathrm{d}t. \tag{7.4}$$

Proof. First, $K - \mathrm{i}/\tau$ maps $D(K)$ onto $\mathcal{H}$. Let $\varphi \in \mathcal{H}$. The function $e^{-t/\tau}U_t\varphi$ is continuous, with the norm $e^{-t/\tau}\,\|\varphi\|$, so $\psi = \int_0^\infty e^{-t/\tau}U_t\varphi\,\mathrm{d}t$ exists. Let h be real. By Lemma 7.7, with U_h taken inside and then the shift $s = t+h$,

$$U_h\psi = \int_0^\infty e^{-t/\tau}U_{t+h}\varphi\,\mathrm{d}t = e^{h/\tau}\int_h^\infty e^{-s/\tau}U_s\varphi\,\mathrm{d}s = e^{h/\tau}\Big(\psi - \int_0^h e^{-s/\tau}U_s\varphi\,\mathrm{d}s\Big).$$

By the last claim of the lemma at $t = 0$, applied on $[-1,1]$ to the function $s \mapsto e^{-s/\tau}U_s\varphi$, which is continuous on all of $\mathbb{R}$, the average $h^{-1}\int_0^h e^{-s/\tau}U_s\varphi\,\mathrm{d}s$ tends to φ. Since also $(e^{h/\tau}-1)/h \to 1/\tau$,

$$\frac{U_h\psi - \psi}{h} = \frac{e^{h/\tau}-1}{h}\,\psi - e^{h/\tau}\,\frac{1}{h}\int_0^h e^{-s/\tau}U_s\varphi\,\mathrm{d}s \longrightarrow \frac{\psi}{\tau} - \varphi.$$

So $\psi \in D(K)$, with $-\mathrm{i}K\psi = \psi/\tau - \varphi$, that is, $(K - \mathrm{i}/\tau)(\mathrm{i}\psi) = \varphi$.

Next, $D(K)$ is dense. Since $\int_0^\infty e^{-t/\tau}\,\mathrm{d}t = \tau$, the vector ψ/τ of the first step, which lies in $D(K)$ for every $\tau > 0$, satisfies

$$\frac{\psi}{\tau} - \varphi = \frac{1}{\tau}\int_0^\infty e^{-t/\tau}(U_t\varphi - \varphi)\,\mathrm{d}t.$$

Split the integral at a time $\delta > 0$. On $[0,\delta]$ the norm of the integrand is at most $e^{-t/\tau}\max_{0\le t\le\delta}\|U_t\varphi-\varphi\|$, and beyond δ at most $2e^{-t/\tau}\,\|\varphi\|$. So the norm bound gives

$$\left\|\frac{\psi}{\tau} - \varphi\right\| \le \max_{0\le t\le\delta}\|U_t\varphi-\varphi\| + 2\,\|\varphi\|\,e^{-\delta/\tau}.$$

By continuity, the first term is small for small δ, and then the second is small for small τ. So every φ is a limit of vectors in $D(K)$, and by Proposition 7.3, K is symmetric.

Then, $K+\mathrm{i}/\tau$ maps $D(K)$ onto $\mathcal{H}$. The family $V_t = U_{-t}$ is a strongly continuous unitary group, and its generator is $-K$, on the same domain, since $(V_h\psi-\psi)/h = -(U_{-h}\psi-\psi)/(-h)$. The first step for (V_t) gives $(-K-\mathrm{i}/\tau)(\mathrm{i}\psi') = \varphi$, with $\psi' = \int_0^\infty e^{-t/\tau}U_{-t}\varphi\,\mathrm{d}t$. That is, $(K+\mathrm{i}/\tau)(-\mathrm{i}\psi') = \varphi$.

Finally, by the test of Proposition 6.7 with $z = \mathrm{i}/\tau$, the operator K is self-adjoint. By Theorem 6.3, i/τ and $-\mathrm{i}/\tau$ lie in its resolvent set, so $K \mp \mathrm{i}/\tau$ is one-to-one, and the vectors $\mathrm{i}\psi$ and $-\mathrm{i}\psi'$ are the two resolvents applied to φ. This is (7.4). ∎

The time τ is arbitrary, and i/τ has the units of K. In words, the resolvent of the generator is the evolution averaged over the future, with weights that fade over the time τ. A group is fixed by its generator.

Proposition 7.9 (A group is fixed by its generator). *Let (U_t) and (V_t) be strongly continuous unitary groups with the same generator K. Then $U_t = V_t$ for every t.*

Proof. Let $\psi \in D(K)$, and put $w(t) = U_t\psi - V_t\psi$. By Proposition 7.3, $w(t) \in D(K)$, and w is differentiable with $w' = -\mathrm{i}Kw$. Divide the identity $\|a\|^2 - \|b\|^2 = \langle a-b, a\rangle + \langle b, a-b\rangle$, for $a = w(t+h)$ and $b = w(t)$, by h, and let $h \to 0$. Since w is differentiable and continuous, and by the symmetry of K,

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\|w(t)\|^2 = \langle w', w\rangle + \langle w, w'\rangle = \mathrm{i}\,\langle Kw, w\rangle - \mathrm{i}\,\langle w, Kw\rangle = 0.$$

Since $w(0) = 0$, $\|w(t)\| = 0$ for every t. So U_t and V_t agree on the dense subspace $D(K)$. Both are bounded, so they agree on every vector. Indeed, if $\psi_n \in D(K)$ tend to χ, then $\|U_t\chi - V_t\chi\| \le 2\,\|\chi - \psi_n\| \to 0$. ∎

Together with the general case, the results of this section match the self-adjoint operators with the evolutions.

Theorem 7.10 (Self-adjoint operators and unitary evolutions · *sketched*). *The map $H \mapsto (e^{-\mathrm{i}tH/\hbar})_{t\in\mathbb{R}}$ is a one-to-one correspondence from the self-adjoint operators onto the strongly continuous unitary groups. Its inverse maps a group to its Hamiltonian $\hbar K$.*

Proof sketch. By Theorem 7.5, sketched, the group $e^{-\mathrm{i}tH/\hbar}$ has the generator $H/\hbar$, so different H give different groups. Let (U_t) be a strongly continuous unitary group with the generator K. By Theorem 7.8, $H = \hbar K$ is self-adjoint, and by Theorem 7.5 the group $e^{-\mathrm{i}tH/\hbar}$ also has the generator K. By Proposition 7.9, $U_t = e^{-\mathrm{i}tH/\hbar}$. ∎

This is the second answer to the question of why observables are self-adjoint. The Hamiltonians of unitary evolutions are exactly the self-adjoint operators. The direction from a group to a self-adjoint generator is proved in full. Only the converse needs the cited spectral theorem, and for the explicit diagonal and multiplication operators it is Theorem 7.4.

The Dirichlet momentum P_D of the box is symmetric, but not self-adjoint (Proposition 3.7). By Stone's theorem, $P_D/\hbar$ is the generator of no strongly continuous unitary group. So P_D generates no unitary evolution, and in particular no translations of the box. A translation in a box needs a rule for what happens at the walls. Lecture 9 builds the translations of the line, which the momentum P generates, and, for each θ, the translations of the box that P_θ generates.

So the three assumptions of § 7.1, a unitary evolution, a law of motion that does not change in time and no jumps, imply by Stone's theorem that the Hamiltonian $\hbar K$ of the evolution is self-adjoint.

Self-check 7.3

1. For the box, compute $\mathrm{i}\int_0^\infty e^{-t/\tau}U_t u_n \,\mathrm{d}t$, and check that it is $(K - \mathrm{i}/\tau)^{-1}u_n$ for the generator $K = H/\hbar$.

Lecture 7, second part

The spreading and narrowing of a free packet

7.4 The free packet

The first part proved that the Hamiltonian of a unitary evolution is self-adjoint, and that the free energy H_0 generates the group (7.3). In this part we follow a free packet under this group. Its mean position moves with a constant velocity, its spread grows at late times, and the chirped Gaussian ψ_b of § 4.5 with $b > 0$ first narrows.

The free evolution determines the position readings at time t. By *Quantum Mechanics I*, Prop. 4.5, the mean of an observable A in the state $U_t\psi$ is the mean of $U_t^\dagger A U_t$ in ψ. On the line, the position is unbounded, so we first find states that the free evolution keeps in $D(Q)$.

Lemma 7.11 (Position in the momentum representation)**.** *Let $\psi \in L^2(\mathbb{R})$ have its transform $\tilde\psi$ in $\mathcal{S}(\mathbb{R})$. Then $\psi \in D(Q) \cap D(P)$, and the transforms of $Q\psi$ and $P\psi$ are $\mathrm{i}\hbar(\tilde\psi)'$ and $p\tilde\psi$, both in $\mathcal{S}(\mathbb{R})$. For every t, the transform of $U_t\psi$ lies in $\mathcal{S}(\mathbb{R})$ as well.*

Proof. The proof of Proposition 4.3, with $e^{+\mathrm{i}px/\hbar}$ in place of $e^{-\mathrm{i}px/\hbar}$, shows that the inverse transform of φ' is $-(\mathrm{i}x/\hbar)F^{-1}\varphi$ for every $\varphi \in \mathcal{S}(\mathbb{R})$. With $\varphi = \tilde\psi$, this gives $x\psi(x) = \mathrm{i}\hbar\,(F^{-1}(\tilde\psi)')(x)$, a square-integrable function. So $\psi \in D(Q)$, and the transform of $Q\psi$ is $\mathrm{i}\hbar(\tilde\psi)'$. Since $p\tilde\psi$ is square-integrable, $\psi \in D(P)$ (Definition 4.4). Both functions lie in $\mathcal{S}(\mathbb{R})$, by the product rule. Finally, each derivative of $e^{-\mathrm{i}tp^2/2m\hbar}$ is a polynomial times $e^{-\mathrm{i}tp^2/2m\hbar}$, so the transform $e^{-\mathrm{i}tp^2/2m\hbar}\tilde\psi$ of $U_t\psi$ lies in $\mathcal{S}(\mathbb{R})$. ∎

On these states, the position acts on the momentum wavefunction $\tilde\psi$ as $\mathrm{i}\hbar\,\mathrm{d}/\mathrm{d}p$. They include the Gaussian packets (4.3), whose transforms Example 4.12 computed.

Proposition 7.12 (The free packet)**.** *Let ψ be a unit vector with $\tilde\psi \in \mathcal{S}(\mathbb{R})$, and with the means $q = \langle Q\rangle$ and $p_0 = \langle P\rangle$. For every real t,*

$$U_{-t}QU_t\psi = Q\psi + \frac{t}{m}P\psi. \tag{7.5}$$

The mean position in $U_t\psi$ is $q + p_0 t/m$, and the spread ΔQ_t of the position in $U_t\psi$ satisfies

$$(\Delta Q_t)^2 = (\Delta Q)^2 + \frac{2t}{m}\,\mathrm{Re}\,\langle (Q-q)\psi, (P-p_0)\psi\rangle + \frac{t^2}{m^2}\,(\Delta P)^2. \tag{7.6}$$

Proof. By Lemma 7.11, applied to ψ and to $U_t\psi$, both lie in $D(Q)\cap D(P)$. By the same lemma and the product rule,

$$(\tilde{QU_t\psi}) = \mathrm{i}\hbar\,\frac{\mathrm{d}}{\mathrm{d}p}\Big(e^{-\mathrm{i}tp^2/2m\hbar}\tilde{\psi}\Big) = e^{-\mathrm{i}tp^2/2m\hbar}\Big(\mathrm{i}\hbar(\tilde{\psi})' + \frac{tp}{m}\,\tilde{\psi}\Big) = e^{-\mathrm{i}tp^2/2m\hbar}\,\Big(Q\psi + \tilde{\frac{t}{m}\,P\psi}\Big).$$

Applying F^{-1} gives $QU_t\psi = U_t(Q\psi + tP\psi/m)$, and applying U_{-t} gives (7.5). Since $U_t^\dagger = U_{-t}$, the mean position is $\langle U_t\psi, QU_t\psi\rangle = \langle\psi, U_{-t}QU_t\psi\rangle = q + p_0 t/m$. With $q_t = q + p_0 t/m$, (7.5) gives $(Q-q_t)U_t\psi = U_t\big((Q-q)\psi + (t/m)(P-p_0)\psi\big)$. Since U_t is unitary, $(\Delta Q_t)^2 = \|(Q-q_t)U_t\psi\|^2 = \|\alpha + c\beta\|^2$, with $\alpha = (Q-q)\psi$, $\beta = (P-p_0)\psi$ and $c = t/m$. Since c is real,

$$\|\alpha + c\beta\|^2 = \|\alpha\|^2 + 2c\,\mathrm{Re}\,\langle\alpha,\beta\rangle + c^2\,\|\beta\|^2\,,$$

and $\|\alpha\| = \Delta Q$ and $\|\beta\| = \Delta P$. This is (7.6). ∎

On these states, $U_{-t}QU_t$ acts as $Q + tP/m$, the position $x + tp/m$ of a classical free particle. By Lemma 0.8, $U_{-t}QU_t$ is self-adjoint, on $U_{-t}D(Q)$. The mean position moves with the constant velocity p_0/m. Twice the real part in (7.6) equals $\langle QP + PQ\rangle - 2\langle Q\rangle\langle P\rangle$, where QP and PQ act on ψ by Lemma 7.11. The variance is a quadratic polynomial in t with the leading coefficient $(\Delta P)^2/m^2$. This coefficient is positive, since $\Delta P = 0$ would make ψ an eigenvector of P, and P has none (Theorem 4.6). So every such packet spreads at late times.

Example 7.13 (The time of flight of a Gaussian packet).

1. *The mixed term.* Take the packet (4.3). By the proof of Theorem 4.9, $(P-p_0)\psi = (\mathrm{i}\hbar/2s^2)(Q-q)\psi$. Since $\|(Q-q)\psi\|^2 = s^2$,

$$\langle (Q-q)\psi, (P-p_0)\psi\rangle = \frac{\mathrm{i}\hbar}{2s^2}\,s^2 = \frac{\mathrm{i}\hbar}{2},$$

   whose real part is 0.

2. *The spread.* With $\Delta Q = s$ and $\Delta P = \hbar/2s$, (7.6) gives

$$(\Delta Q_t)^2 = s^2 + \frac{\hbar^2 t^2}{4m^2 s^2} = s^2\Big(1 + \frac{t^2}{t_s^2}\Big), \qquad t_s = \frac{2ms^2}{\hbar}.$$

   For $|t| \ll t_s$, much less than the *spreading time* t_s, the width hardly changes, and for $t \gg t_s$ it grows as $(\Delta P/m)\,t$.

3. *An electron and an atom.* For the electron of Example 4.12, with $s = 0.1\,\mathrm{nm}$, the spreading time is $t_s \approx 1.7\times10^{-16}\,\mathrm{s}$. For a rubidium-87 atom, of mass $1.443\times10^{-25}\,\mathrm{kg}$, with $s = 1\,\mu\mathrm{m}$, it is $t_s \approx 2.7\,\mathrm{ms}$, and the velocity spread is $\hbar/2ms \approx 0.37\,\mathrm{mm/s}$. After 20 ms of free flight, $\Delta Q_t \approx 7.4\,\mu\mathrm{m}$, close to $(\Delta P/m)\,t \approx 7.3\,\mu\mathrm{m}$.

A packet can also narrow before it spreads, when the mixed term in (7.6) is negative. Take the chirped Gaussian ψ_b of § 4.5, whose transform is again a Gaussian and so lies in $\mathcal{S}(\mathbb{R})$. Indeed, let $c = (1+\mathrm{i}b)/4s^2$. The first step of Example 4.12 applies to $G_b(k) = \int e^{-\mathrm{i}ky}e^{-cy^2}\,\mathrm{d}y$ with the same Taylor bound, since $\big|e^{-cy^2}\big| = e^{-y^2/4s^2}$. It gives $G_b'(k) = -(k/2c)G_b(k)$. So

$G_b(k) = G_b(0)e^{-k^2/4c}$, and $\left|e^{-k^2/4c}\right| = e^{-s^2k^2/(1+b^2)}$. The transform of ψ_b is a constant times $G_b(p/\hbar)$.

Since $\psi_b' = -(1+\mathrm{i}b)x\psi_b/2s^2$,

$$P\psi_b = -\mathrm{i}\hbar\psi_b' = \frac{\mathrm{i}\hbar(1+\mathrm{i}b)}{2s^2}\,Q\psi_b.$$

The density $|\psi_b|^2$ is even, so $\langle Q\rangle = 0$, and by this relation $\langle P\rangle = 0$. Since $\|Q\psi_b\|^2 = s^2$,

$$\langle Q\psi_b, P\psi_b\rangle = \frac{\mathrm{i}\hbar}{2} - \frac{\hbar b}{2} \quad \text{and} \quad (\Delta P)^2 = \|P\psi_b\|^2 = \frac{\hbar^2(1+b^2)}{4s^2}.$$

So $\operatorname{Re}\langle Q\psi_b, P\psi_b\rangle = -\hbar b/2$, and (7.6) gives

$$(\Delta Q_t)^2 = s^2 - \frac{\hbar b t}{m} + \frac{\hbar^2(1+b^2)t^2}{4m^2s^2}.$$

This quadratic polynomial in t has its minimum at $t_* = 2ms^2b/\hbar(1+b^2)$. So for $b > 0$ the packet narrows until the time t_*, where the variance is $s^2/(1+b^2)$, and then it spreads. At t_*, $\Delta Q_t\,\Delta P = \hbar/2$, the lower bound of the uncertainty relation.

Figure 7.1 compares this with the unchirped packet, for $b = 1$. The chirped packet is narrowest, with $\Delta Q_t = s/\sqrt{2}$, at $t = t_s/2$. The two spreads are equal at $t = 2t_s$, and after that the chirped packet is the wider one, since its ΔP is larger.

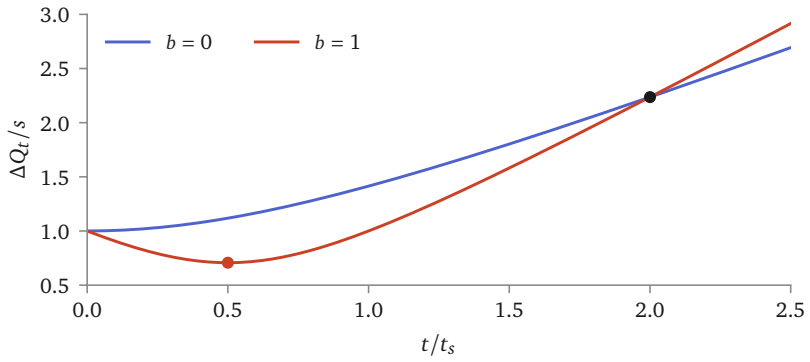


Figure 7.1: Position spreads for an unchirped packet ($b = 0$) and the chirped packet of § 4.5 with $b = 1$, both initially of width s. Time is in units of $t_s = 2ms^2/\hbar$; the mixed term lets the latter narrow first. The dots mark the minimum of the curve for $b = 1$ and the point where the curves meet.

The mixed term is a correlation between position and momentum. For a unit vector ψ as in Proposition 7.12, with $q_t = q + p_0t/m$, let

$$C_t = \operatorname{Re}\langle (Q-q_t)U_t\psi, (P-p_0)U_t\psi\rangle\,,$$

so that C_0 is the real part in (7.6). The momentum acts on the transform by multiplication, so $(P-p_0)U_t\psi = U_t(P-p_0)\psi$. With the formula for $(Q-q_t)U_t\psi$ in the proof of Proposition 7.12, this gives

$$C_t = C_0 + \frac{t}{m}\,(\Delta P)^2,$$

and (7.6) says that $(\Delta Q_t)^2$ has the derivative $2C_t/m$. So the spread falls while $C_t < 0$ and grows while $C_t > 0$. The chirped Gaussian with $b > 0$ starts with $C_0 = -\hbar b/2 < 0$. The

expression (2.5), applied to ψ_b, is $j = -(\hbar b x/2ms^2)\,|\psi_b|^2$. Figure 7.2 draws this current for $b = 1$ at $t = 0$, with the densities at $t = 0$ and at the narrowest time $t_* = t_s/2$. The arrows point towards the centre from both sides. Taken as a flow of probability, the current moves probability towards the centre, and the packet narrows. This interpretation of j is not proved on the line, since Lecture 2 proved it only for finite combinations of the box states.

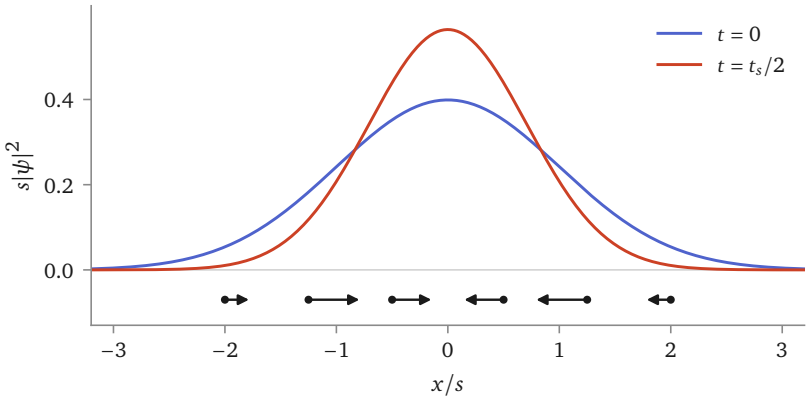


Figure 7.2: The chirped Gaussian ψ_b with $b = 1$, in units with $\hbar = m = s = 1$. The curves are its position densities at $t = 0$ and at $t = t_s/2$, where it is narrowest. Each arrow below the axis starts at a point x, points in the direction of the current $j(x,0) = -x\,|\psi_b(x)|^2/2$ there, and has a length proportional to $|j(x,0)|$. The arrows show a flow of probability, not the paths of particles.

The free evolution also makes chirps. The packet (4.3) with $q = p_0 = 0$, evolved for a time t, is the chirped Gaussian with the width $s(1 + t^2/t_s^2)^{1/2}$ and $b = -t/t_s$, up to a constant phase (Teta, (6.72)). For a Gaussian under free evolution, the quadratic phase expresses this correlation. This packet has $C_0 = 0$, so $C_t = t(\Delta P)^2/m = -\hbar b/2$, with $b = -t/t_s$.

The second moments of a packet as in Proposition 7.12 form its *covariance matrix*

$$\Sigma_t = \begin{pmatrix} (\Delta Q_t)^2 & C_t \\ C_t & (\Delta P)^2 \end{pmatrix}.$$

By (7.6) and the formula for C_t,

$$\Sigma_t = S_t \Sigma_0 S_t^{\mathsf{T}}, \qquad S_t = \begin{pmatrix} 1 & t/m \\ 0 & 1 \end{pmatrix},$$

where S_t is the matrix of the free motion $(x, p) \mapsto (x + tp/m, p)$ of a classical particle. So the second moments change as those of a cloud of classical free particles, in which the faster particles run ahead.

For the chirped Gaussian,

$$\det \Sigma_0 = s^2\,\frac{\hbar^2(1+b^2)}{4s^2} - \frac{\hbar^2 b^2}{4} = \frac{\hbar^2}{4},$$

and $\det \Sigma_t = \det \Sigma_0$, since $\det S_t = 1$. Since also $(\Delta Q_t)^2 > 0$, the matrix Σ_t is positive definite. Its *covariance ellipse* at the time t is the set of the points (x, p) with $z^{\mathsf{T}} \Sigma_t^{-1} z = 1$, where $z = (x - q_t,\ p - p_0)$. For z on the ellipse and $e = (1, 0)$, the Cauchy–Schwarz inequality for the inner product $u^{\mathsf{T}} \Sigma_t^{-1} v$ gives

$$e^{\mathsf{T}} z = (\Sigma_t e)^{\mathsf{T}} \Sigma_t^{-1} z \le \left(e^{\mathsf{T}} \Sigma_t e\right)^{1/2} = \Delta Q_t,$$

with equality at $z = \Sigma_t e/\Delta Q_t$. So the shadow of the ellipse on the x axis is the interval from $-\Delta Q_t$ to ΔQ_t. Since $\Sigma_t = S_t \Sigma_0 S_t^{\mathsf{T}}$, the ellipse at the time t is the image of the ellipse at the time 0 under the free motion. The ellipse records second moments only. It is not a contour of a joint distribution of position and momentum readings.

Figure 7.3 draws the covariance ellipses of ψ_b with $b = 1$. At $t = 0$ the ellipse is tilted down to the right, since $C_0 < 0$. At $t = t_s/2$ we have $C_t = 0$, so Σ_t is diagonal and the axes of the ellipse lie along the coordinate axes. In the dimensionless coordinates of the figure it is the circle of radius $1/\sqrt{2}$, and its shadow is shortest. After that it tilts up to the right, as a cloud of free particles does when its faster particles run ahead, and position and momentum become positively correlated.

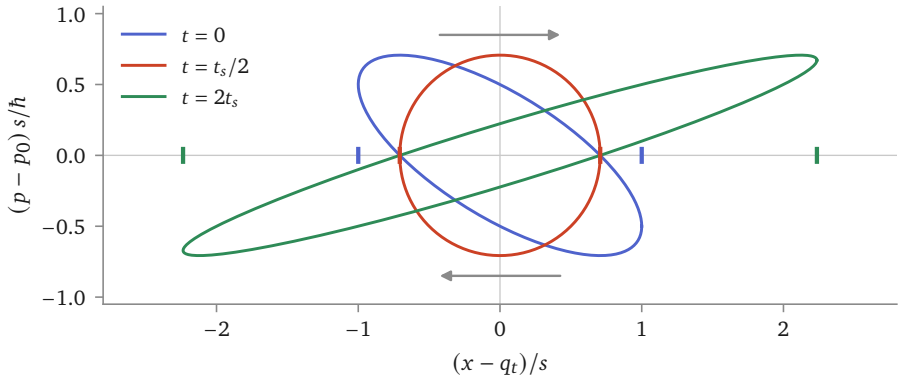


Figure 7.3: Covariance ellipses of the chirped Gaussian ψ_b with $b = 1$, at $t = 0$, $t = t_s/2$ and $t = 2t_s$, in units with $\hbar = m = s = 1$. The short marks on the horizontal axis show $\pm\Delta Q_t$, in the colour of each ellipse. The grey arrows show the free motion $(x, p) \mapsto (x + tp/m, p)$ for $t = 1$ at two momenta.

Now read the formula for the evolved packet backwards, with the start width $s_0 = s/\sqrt{1 + b^2}$ and its spreading time $t_{s_0} = 2ms_0^2/\hbar$. Then $t_* = bt_{s_0}$, and at the time $-t_*$ the formula gives the width s and the chirp b. So ψ_b with $b > 0$ is, up to a constant phase, the packet (4.3) with $q = p_0 = 0$ and the width s_0, evolved for the time $-t_*$. So the narrowing is the evolution of such a packet at negative times.

By (7.6), $\Delta Q_t/t \to \Delta P/m$ as $t \to \infty$, for every such packet. At late times the spread of the positions is the spread of the velocities times t. By a cited theorem, Lecture 12 shows more. The probabilities of Q/t on intervals tend to those of the velocity P/m.

Self-check 7.4

1. An electron starts in the packet (4.3) with $s = 0.1$ nm. After how long has its spread doubled?

The physics

The temperature of a cold gas is measured by letting it fly. In 1988 Lett, Watts, Westbrook, Phillips, Gould and Metcalf cooled sodium atoms with laser light, switched the light off, and timed the atoms' arrival at a probe beam below. The arrival times gave the spread of the velocities, and so the temperature: about 43 μK, below the limit of about 240 μK that the theory of laser cooling then predicted. At 43 μK the velocity spread $\sqrt{k_B T/m}$ of a sodium atom is about 0.12 m/s, so in 10 ms of flight these velocities alone displace an atom by about 1.2 mm, as an rms distance. The cloud holds many atoms and falls under gravity, so the free packet of this lecture is a model of

one atom. For one free atom, (7.6) says that after a long flight its spread grows as its velocity spread times the time.

Dynamics

Evolution: the generator of every strongly continuous unitary group is self-adjoint (Theorem 7.8), and every self-adjoint H generates the group $e^{-\mathrm{i}tH/\hbar}$, proved for the explicit diagonal and multiplication operators (Theorem 7.4) and sketched in general (Theorem 7.5). The Schrödinger equation holds exactly on $D(H)$. *Observables:* the Hamiltonians of unitary evolutions are the self-adjoint operators (Theorem 7.10, sketched), and the Dirichlet momentum generates no unitary group. *Measurement:* a free particle's momentum density never changes. For a state with a rapidly decreasing transform, the variance of its position is quadratic in t (Proposition 7.12), and at late times its spread grows as $\Delta P\, t/m$.

Summary.

- *Unitary groups and their generators* (Definitions 7.1 and 7.2): $U_t\psi$ is differentiable exactly on $D(K)$, where $\mathrm{i}\,\mathrm{d}U_t\psi/\mathrm{d}t = KU_t\psi$ (Proposition 7.3).
- *Evolution by an explicit Hamiltonian*: diagonal operators and multiplication by a real continuous function, after a unitary, generate groups (Theorem 7.4); every self-adjoint operator does, by a sketched proof (Theorem 7.5).
- *Stone's theorem*: the generator is densely defined and self-adjoint, with the resolvents at $\pm\mathrm{i}/\tau$ as Laplace integrals of the group (Theorem 7.8); a group is fixed by its generator (Proposition 7.9); self-adjoint operators and unitary evolutions correspond one to one, by a sketched proof (Theorem 7.10).
- *The free packet*: if $\tilde{\psi} \in \mathcal{S}(\mathbb{R})$, then $U_{-t}QU_t\psi = Q\psi + tP\psi/m$, and the position variance is quadratic in t (Proposition 7.12); a Gaussian's spreading time is $t_s = 2ms^2/\hbar$ (Example 7.13), and a negative correlation of position and momentum makes a chirped Gaussian narrow first.

Lecture 8 evolves the harmonic oscillator, whose group returns every state, up to a common sign, after one period.

Self-check answers.

7.1 1. Each U_t is unitary, $U_0 = \mathbb{1}$, $e^{-\mathrm{i}c(t+s)} = e^{-\mathrm{i}ct}e^{-\mathrm{i}cs}$, and $\|U_t\psi - \psi\| = \left|e^{-\mathrm{i}ct} - 1\right| \|\psi\| \to 0$. Since $\mathrm{i}(e^{-\mathrm{i}ch} - 1)/h \to c$, every vector lies in $D(K)$, and $K = c\mathbb{1}$.

7.2 1. Yes, since $\int p^2/(1+p^2)^2\,\mathrm{d}p = \pi/2$ is finite. No, since $p^2/(1+p^2) \to 1$, so $p^2\tilde{\psi}$ is not square-integrable. By Theorem 7.4 and Proposition 7.3, $t \mapsto U_t\psi$ is differentiable at no t.

7.3 1. Since $U_t u_n = e^{-\mathrm{i}E_n t/\hbar}u_n$, the integral is $\mathrm{i}u_n/(1/\tau + \mathrm{i}E_n/\hbar) = u_n/(E_n/\hbar - \mathrm{i}/\tau)$, and $(K - \mathrm{i}/\tau)u_n = (E_n/\hbar - \mathrm{i}/\tau)u_n$.

7.4 1. $\Delta Q_t = 2s$ when $1 + t^2/t_s^2 = 4$, at $t = \sqrt{3}\,t_s \approx 3.0 \times 10^{-16}$ s.

References

Other books prove some results of this lecture. Stone's theorem with the resolvents (7.4) is Amrein, Prop. 5.1 and Cor. 5.2, with $z = \mathrm{i}/\tau$. Hall, Thm 10.15, and Teschl, Thm 5.2,

prove Stone's theorem in another way. Teta, (6.70), writes twice the real part in (7.6) as $\langle QP + PQ\rangle - 2\langle Q\rangle\langle P\rangle$.

Hall, *Quantum Theory for Mathematicians*, Def. 10.5 (p. 206), Thm 10.10 and Def. 10.11 (p. 207), Prop. 10.14 (pp. 208–209), Thm 10.15 (p. 210). Amrein, *Hilbert Space Methods in Quantum Mechanics*, §5.1.1 (p. 193), Prop. 5.1 and Cor. 5.2 (pp. 194–196). Teschl, *Mathematical Methods in Quantum Mechanics*, Thm 5.1 (p. 123), Thm 5.2 (p. 124). Teta, *A Mathematical Primer on Quantum Mechanics*, (6.70) (p. 150), (6.72) (p. 151). *Quantum Mechanics I*, Thm 4.4 and Prop. 4.5. Lett, Watts, Westbrook, Phillips, Gould and Metcalf, *Phys. Rev. Lett.* 61, 169 (1988).